

RARE: Decoupling Representation Steering from Expert Routing in Mixture-of-Experts Language Models

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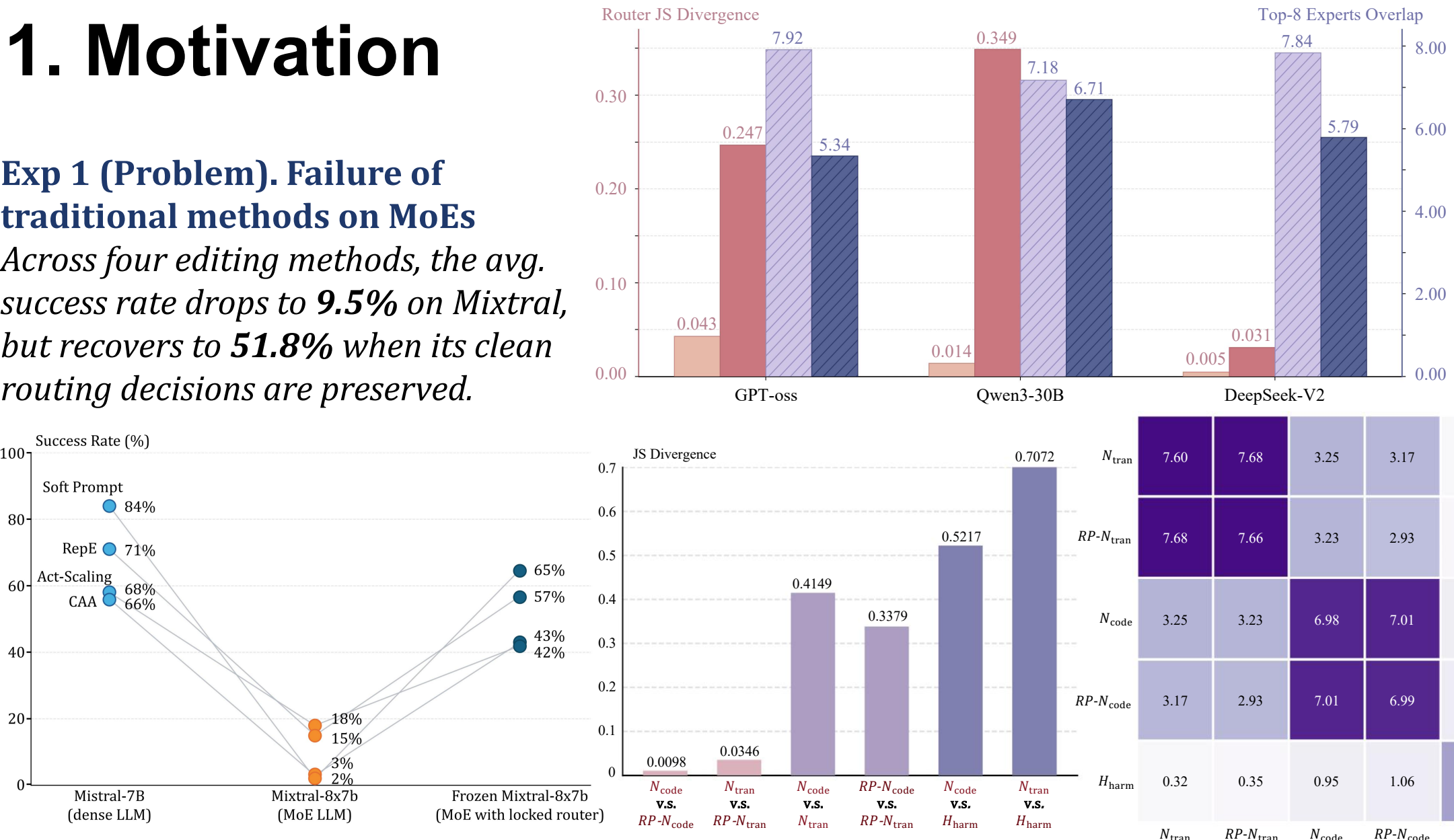
**Problem**
— Empirical studies show why router steering fails on MoE models.

**Idea**
— RARE: Router-Agnostic Representation Engineering

**Result**
— RARE improves steering effectiveness across 6 MoEs and 3 tasks

1. Motivation

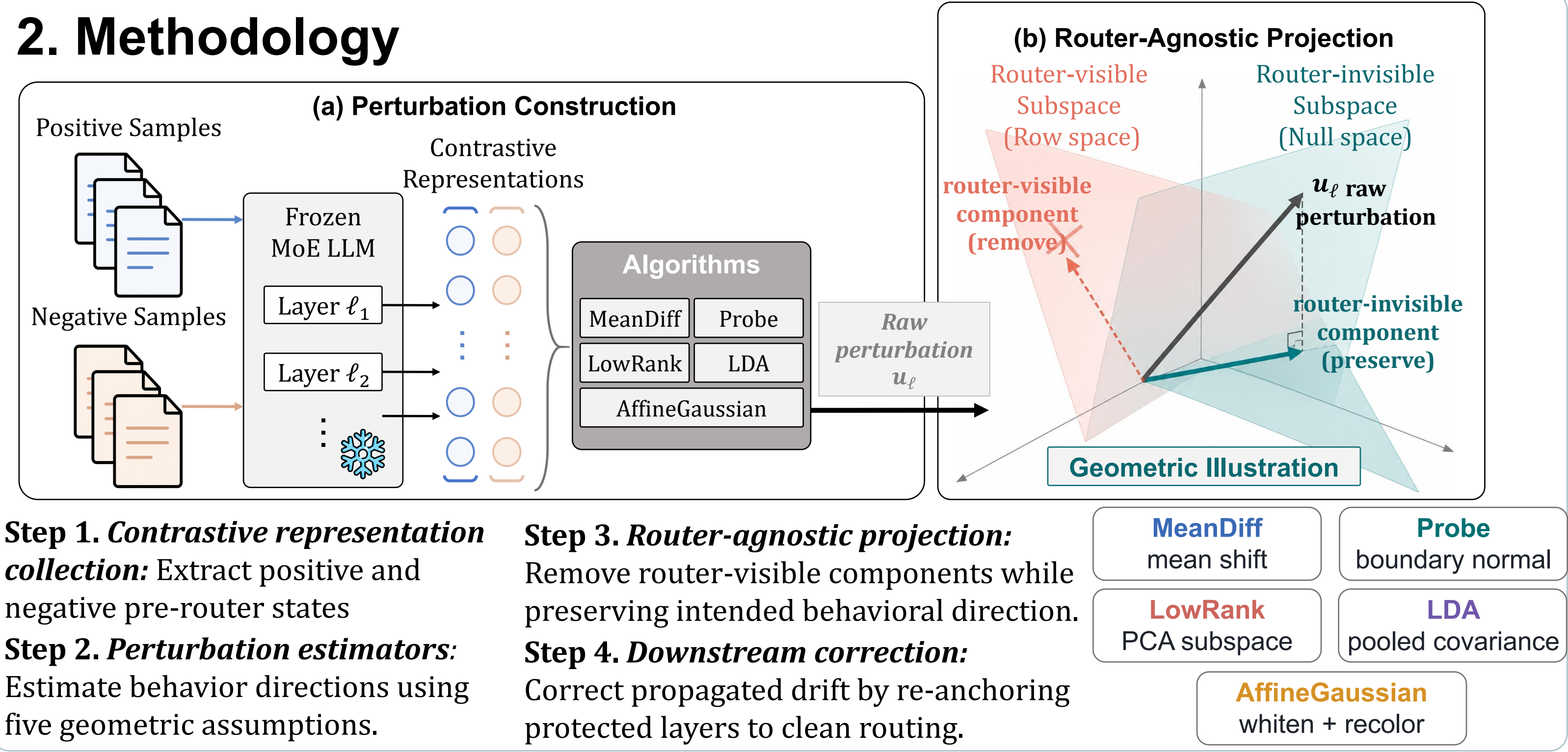
Exp 1 (Problem). Failure of traditional methods on MoEs
Across four editing methods, the avg. success rate drops to **9.5%** on Mixtral, but recovers to **51.8%** when its clean routing decisions are preserved.



Exp 2 (Finding 1). Routing similarity between different response types.
Lighter: refusal and compliant responses of the **same** harmful query ;
Darker: refusals of harmful queries of **different** topics.
Same-query **refusal/compliance** pairs exhibit lower router JSD and higher Top-8 expert overlap than the cross-query refusal controls.

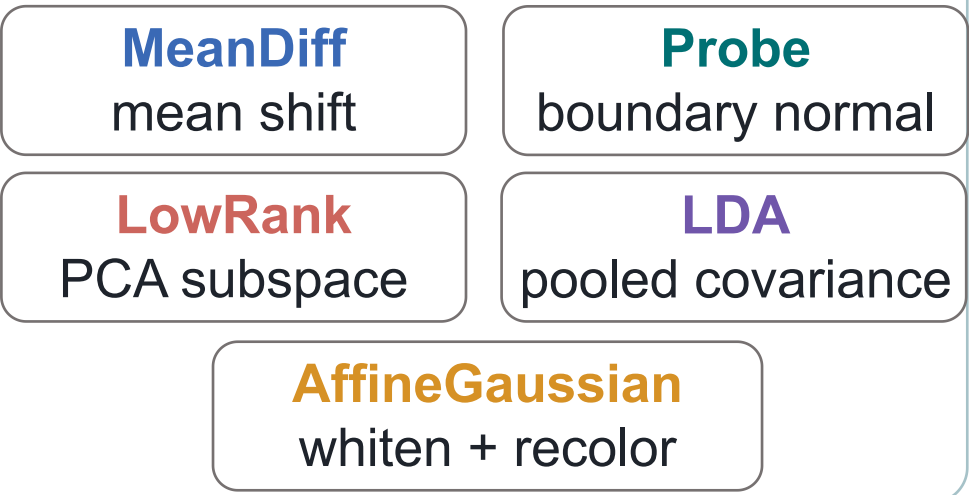
Exp 3 (Finding 2). Query-level routing comparisons.
Left: pairwise router-logits JSD; **Right:** Top-8 expert overlap.
Original and refusal-inducing queries within the same semantic domain retain similar routing, whereas cross-domain and benign-harmful comparisons produce substantially larger routing shifts.

2. Methodology



Step 1. Contrastive representation collection: Extract positive and negative pre-router states
Step 2. Perturbation estimators: Estimate behavior directions using five geometric assumptions.

Step 3. Router-agnostic projection: Remove router-visible components while preserving intended behavioral direction.
Step 4. Downstream correction: Correct propagated drift by re-anchoring protected layers to clean routing.



4. Conclusion / Takeaways

✓ Routing consistency is essential for effective representation steering in MoE models.

✓ RARE decouples behavior control from expert allocation by removing router-visible perturbation components.

✓ Across 6 MoEs and 3 tasks, RARE (with AffineGaussian) improves steering in all 18 model-task settings.

QR code for paper link:



3. Evaluation

6 MoE backbones | 3 benchmarks & metrics | 2 baselines | 3 tasks

Task-specific evaluation

- **Harmfulness steering:** JailbreakBench and MaliciousInstruct
Metrics: Attack Success Rate (ASR) and MMLU accuracy
- **Truthfulness steering:** TruthfulQA **Metric:** MC1 accuracy
- **Factual editing:** CounterFact
Metrics: Efficacy Score (ES) and Locality Score (LS)

53.3%
Harmfulness ASR
Clean **8.9** | RepE **18.7** | SteerMoE **46.3**
RARE (AffineGaussian) **53.3**

67.8%
MMLU capability
RARE (AffineGaussian) **67.8** | RepE **65.7** | SteerMoE **64.5** | Clean **77.8 (-10.0)**

58.6%
TruthfulQA MC1
RARE (AffineGaussian) **58.6** | RepE **42.2** | Clean **41.0 (+17.6)**

96.3%
CounterFact efficacy
RARE (AffineGaussian) **96.3** | RARE (LDA) **39.5** | RepE **28.8** | Clean **16.8 (+79.5)**

Projection ablation

- Direct AffineGaussian: **14.6%** average ASR
- AffineGaussian + RARE projection and correction: **53.3%**
- Improvement: **+38.7** percentage points.

Summary: Router-agnostic projection is the key component that makes representation steering effective across MoE architectures and behavioral tasks.